Reza Sajjadinasab

PhD Candidate, Computer Engineering | Efficient Training & Inference, Post-Training Foundation Models

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SUMMARY

Computer engineering PhD candidate who makes post-training of large pretrained models cheap and fast. I designed and built Greg, which post-trains a 361M-parameter vision model on a frozen DINOv3 backbone with RL and a learned latent world model, raising its Bench2Drive driving score from 84.28 to 92.03, comparable to the strongest published camera-only policy, while updating only 7 layers (full fine-tuning reaches 91.21), training on two consumer GPUs, and running at 27.4 FPS. With **Red Hat**, I build learned optimizers over code-LLM embeddings. Looking to contribute to efficient training and inference of diffusion language models and multimodal generative models. Graduating August 2027.

EDUCATION

Ph.D., Computer Engineering, Boston University — Co-advisors: Profs. Herbordt & Ohn-Bar — Boston, MA | 2022 – Aug 2027 (expected)

M.S., Computer Engineering, Boston University — Advisor: Prof. Martin Herbordt — Boston, MA | 2022 – 2024

B.S., Electrical Engineering, Sharif University of Technology — Advisor: Prof. Iman Gholampour — Tehran, Iran | 2017 – 2022

RESEARCH EXPERIENCE

Graduate Researcher — Reinforcement Learning for Autonomous Driving — **Boston University**

Advisor: Prof. Eshed Ohn-Bar | PyTorch, CARLA, HUGSIM — Boston, MA | 2024 – Present

- Designed and built **Greg**, off-policy RL fine-tuning of an **imitation-learned (IL)** camera-only policy posed as a **constrained MDP**: SAC with an augmented-Lagrangian cost budget and a KL anchor to the IL policy, **distributional critic ensembles** whose uncertainty prioritizes replay, and **value expansion from a latent world model** trained without RL gradients (under review).
- Raised CARLA Bench2Drive driving score from **84.28 to 92.03**, comparable to the strongest published camera-only policy (91.60), and improved every IL initialization tested (**30.47** $\rightarrow$ **60.38**, **69.19** $\rightarrow$ **78.11**).
- **Generalized beyond the demonstrations**: 54.26 driving score with **zero collisions** on 28 adversarial routes vs 27.96 for the strongest IL baseline; **zero-shot transfer** to HUGSIM, reconstructed from real captures (HD-Score 0.30 vs 0.16; 0.36 after 12 h of in-domain RL).
- **Training recipes**: vanilla SAC drops 30.8 points below the IL policy, while the full recipe stays above it at every budget and holds 76.9 at batch 64 where PPO falls to 20.7; a geometric-mean reward beats an additive one by 23.8; 10 critics beat 2 by 8.2.
- **Architecture & efficiency**: frozen DINOv3 encoder over six cameras with a transformer head (361M parameters, 303M frozen); post-training 7 layers matches full fine-tuning (92.03 vs 91.21) at lower memory and compute; **27.4 FPS** on one RTX 3090, **100 FPS** for a lighter direct-control model.
- Built the **asynchronous actor–learner system** on 2$\times$ RTX 3090: simulators on one GPU feed a shared replay buffer while the learner trains on the other at a fixed 3 updates per environment step; 480 GPU hours per run.

Graduate Researcher — ML-Driven Compiler Optimization — **Boston University | Red Hat**

Advisor: Prof. Martin Herbordt | Mentors: Sanjay Arora and Ulrich Drepper, Red Hat — Boston, MA | 2022 – Present

- Built **Flagwise**, a learned model over code-LM embeddings that acts when every action costs a verified build: its **first pick, made before any feedback, is 1.03% smaller than `-Oz`** where every learned ranker we tested is larger, and model-guided search matches in 20 builds what random search needs 169 for (under review); earlier, compared RL, Bayesian optimization, and genetic search (IEEE HPEC 2024).

SELECTED PUBLICATIONS & HONORS

- **R. Sajjadinasab**, E. Ohn-Bar. "Greg: A Guided Recipe for Efficient Reinforcement Fine-Tuning in Sensorimotor Driving." *Under review*, 2026.
- **R. Sajjadinasab**, S. Arora, M. Herbordt. "Learned Compiler Flag Selection That Beats the Default from the First Build." *Under review*, 2026.
- **R. Sajjadinasab**, S. Arora, U. Drepper, A. Sanaullah, M. Herbordt. "A Graph-Based Algorithm for Optimizing GCC Compiler Flag Settings." *IEEE HPEC*, 2024.
- **8 papers** in total (4 published, 4 under review), including IEEE IPDPSW '24, *IET ITS* '22, and *ASCE JTE* '24. Google Scholar
- Distinguished Computer Engineering Fellowship, Boston University, 2022; ranked **192nd of** $\sim$**200,000** in Iran's national university entrance exam (top 0.1%), 2017

TECHNICAL SKILLS

Reinforcement & Imitation Learning SAC, PPO, constrained MDPs (augmented Lagrangian), distributional critic ensembles, prioritized replay, model-based value expansion, latent world models (RSSM), behavior cloning, reward design, KL-regularized post-training, zero-shot transfer

Deep Learning PyTorch, NumPy; frozen foundation encoders with adapters (DINOv3 + ViT-Split), transformer heads, multi-camera inputs, hybrid discrete/continuous action heads, distributed asynchronous training

Inference & Scale (working knowledge) KV-cache sizing, weight and KV-cache quantization (fp8/int4), continuous batching, mixed precision; data/tensor/pipeline parallelism, FSDP/ZeRO

Languages & Tools Python, C/C++, CUDA, CARLA, Bench2Drive, HUGSIM, Linux, Git